Aurelio Losquiño Muñoz

Artificial vision engineer

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github | linkedin

Education

Autonomous University of Barcelona, Spain *Sep. 2016 - Jun. 2020*

Degree in Mathematics

Specialization in Fundamental Mathematics

University of Barcelona, Spain *Sep. 2020 - Sep. 2021*

Master's Degree in Advanced Mathematics

Specialization in Computational Algebra

Experience

Private Tutor for University Students *2019 - 2021*

Mathematics and Programming Tutor for Engineering Students

- Assisted four engineering students from different universities such as **UPC, UB,** and **laSalle**.
- Subjects taught: **C Programming** and **Multivariable Calculus**.

Smalle Technologies *Apr. 2020 - Sep. 2021*

Software Developer Trainee

- Developed **web scraping** scripts to obtain datasets about water quality in Catalonia.
- Trained **machine learning models** to predict water quality based on cheaper variables to measure.
- Programmed the backend and a **NodeJS+MySQL REST API** for processing, storing, and delivering sensor data deployed in fish farms.

Smalle Technologies *Sep. 2021 - Oct. 2023*

Artificial Vision Engineer

- Trained models using a **custom detection dataset** with YOLOv4, MobileNet, **YOLOv7,** and **YOLOv8** architectures.
- Trained **instance segmentation models** using **Mask RCNN** and **YOLOv7-seg** architectures.
- Optimized models in **TensoRT** format with **FP16** precision for **real-time inference**.
- Developed complete pipelines for deploying **artificial vision applications with GStreamer and Deepstream** on **Nvidia Jetson** devices.

- Developed software capable of **counting and estimating the mass of each fish in real-time** at a speed of 800 fish/min.
- Developed software capable of **detecting dead or unhealthy fish** and providing alerts for operators.
- These software solutions were commercialized and implemented in several fish farms, generating substantial revenue for the company.

ALSTOM *Feb. 2024 - Present*

COMPUTER VISION ENGINEER

- Developed a Python library used for **3D camera calibration and 3D reconstruction**. This library required a deep knowledge of **linear algebra, projective geometry, and computer vision techniques**.
- Developed a software to retrieve results from a 3D train reconstruction analysis and publish it to client visualization app.
- Developed a software for a R&D 3D vision system measure train wheels diamter with an error lower that 0.5mm.
- Developed a software to compare 2D images of a same object taken at different times to detect and measure deformations in the object.
- Manage git+gitlab and docker for the software team.

SKILLS

LANGUAGES	***Native:*** Spanish | Catalan ***Professional:*** English
PROGRAMMING LANGUAGES	Python | C | JavaScript/NodeJS | Java
V.A. FRAMEWORKS / LIBRARIES	YOLO models | GStreamer | Deepstream | PyTorch | Docker
DATA FRAMEWORKS/LIBRARIES	Pandas | scikit-learn | Matplotlib
CLOUD	Google Cloud | Google App Engine | Heroku | AWS
SOFT SKILLS	Strong Problem Solving | Autonomy | Good Communicator | Rigor

PERSONAL PROJECTS

- Implemented the **Minimax algorithm with alpha-beta pruning** in **C language**, optimized with variations of the **iterative deepening** algorithm for the games of **chess, checkers,** and **connect four**.
- Developed a **Python and Selenium bot** for the website playok.com to **automate moves**. The program easily surpassed the level of an average player. https://github.com/aurelm95/DAMAS-playok-Bot
- Created a **Python bot** for the online game https://www.travian.com/es with **telegram alerts**, allowing me to progress in the game at an nonhuman rate.
- Implemented a **neural network from scratch** in Python using the **backpropagation algorithm** to train the network to **recognize handwritten digits** (MNIST). https://NumberRecognition.aurelm.repl.co
- Implemented the **"NeuroEvolution of Augmenting Topologies" (NEAT) genetic algorithm**. It was used in various **reinforcement learning environments**.

- Developed software to find **sure bets using web scraping** with Python requests from different **betting websites** (bet365, williamhill, bwin...).
- Tracked, stored, and visualized price variations by **daily web scraping from Amazon**.

Collaborations

- Collaborated with the "Centre de Recerca Matemàtica" (CRM) by mentoring a high school final project on **mathematics and artificial intelligence applied to chess**.

Open Source Contributions

- Contributed to the **state-of-the-art repository** https://github.com/ultralytics/ultralytics (**YOLOv8**) by generating the **Dockerfile** for **Nvidia Jetson** devices.
- Contributed to the repository https://github.com/aurelm95/yolov7-seg by **fixing a bug** that prevented proper model export.

Other Interests

- Chess, traveling, and sports in general.